5.3 Recursive Definitions and Structural Induction

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Sometimes the most convenient way to define a set, function, sequence etc. is by using itself.

This includes the Fibonacci sequence and even $\{2^n\}_{n=0}^{\infty}$ which can be thought of as $a_0=1$, $a_n=2a_{n-1}$. Using these recursive definitions allows us to use induction to prove results about the sequence, set, function.

Recursively Defined Functions

- Define f at 0 (or some b).
- Give a rule for finding $f(n)$ using smaller values (anything in $\{0,1,\dots,n-1\}$).

Let f be the function defined by

$f(0)=3$, $f(n+1)=2f(n)+3$.

$$f(1)\overset{?}{=}2f(0)+3=2\cdot3+3=9$$
$$f(2)\overset{?}{=}2f(1)+3=2\cdot9+3=21$$
$$f(3)\overset{?}{=}2f(2)+3=2(21)+3=45$$
$$f(4)\overset{?}{=}2f(3)+3=93$$

Give a recursive definition of a^n where $a\in\mathbb{R}$ $a\neq0$ and $n\in\mathbb{N}$.

$$a^0=1,\quad a^{n+1}=a\cdot a^n.$$

Give a recursive definition of $\sum_{i=0}^{n}a_i$.

$$\sum_{i=0}^{0}a_i=a_0,\quad\text{and}$$

$$\sum_{i=0}^{n+1}a_i=\sum_{i=0}^{n}a_i+a_{n+1}.$$

Let S be the set defined as

$3\in S$, and if $x,y\in S$ $x+y\in S$.

$3\in S$ so $3+3=6\in S$

$3,6\in S$ so $3+6=9\in S$

$6,6\in S$ so $6+6=12\in S$

S contains $\{3,6,9,12,\dots\}$

$S=\{x\mid(x=3n)\wedge(n\in\mathbb{Z}^+)\}$

Let $f(0)=3$, find $f(2)$ if

1) $f(n+1)=-2f(n)$ $\quad f(1)=-2(3)=-6,\ f(2)=-2(-6)=12$
2) $f(n+1)=f(n)^2-2f(n)-2$ $\quad f(1)=3^2-2\cdot3-2=1,\ f(2)=1^2-2-2=-3$
3) $f(n+1)=3^{f(n)/3}$ $\quad f(1)=3^{3/3}=3,\ f(2)=3^{3/3}=3$

Recursively define the sequences

1) $\{6n\}_{n=1}^{\infty}$ $\quad a_1=6 \quad a_n=a_{n-1}+6$
2) $\{n(n+1)\}_{n=1}^{\infty}$ $\quad a_1=2 \quad a_n=a_{n-1}+2n$ $\quad$ Compare this to n^2

Show that whenever $n\geq3$, $f_n>\alpha^{n-2}$ where $\alpha=(1+\sqrt{5})/2$ and f_n is the n^{th} term of the Fibonacci sequence.

Let $\alpha=(1+\sqrt{5})/2$, $\{f_n\}_{n=0}^{\infty}$ be the Fibonacci sequence, and $P(n)$: "$f_n>\alpha^{n-2}$" where n is an integer greater than or equal to 3.

Base Case ($n=3;4$) $P(3)$: "$f_3>\alpha^{3-2}$" where $f_3=f_2+f_1=2$ and $\alpha^{3-2}=\left(\frac{1+\sqrt{5}}{2}\right)^1$ but $1+\sqrt{5}<1+3$ so $\frac{1+\sqrt{5}}{2}<\frac{4}{2}=2$ hence $\alpha<f_3$ and $P(3)$ is true. $P(4)$?

Strong Inductive Step: Assume $P(i)$: "$f_i<\alpha^{i-2}$" is true for every $4\leq i\leq k$ for an arbitrary integer $k\geq4$. We want to show $P(k+1)$: "$f_{k+1}<\alpha^{k-1}$" is true. Notice

$$f_{k+1}=f_k+f_{k-1}<\alpha^{k-2}+\alpha^{k-3}\text{ by}$$

$P(k)$ and $P(k-1)$.

Need $\alpha^{k-1}\overset{?}{\geq}\alpha^{k-2}+\alpha^{k-3}\Rightarrow\alpha^{k-1}\overset{?}{\geq}(\alpha+1)\alpha^{k-3}$

$\Rightarrow\alpha^2\overset{?}{\geq}(\alpha+1)\Rightarrow\left(\frac{1+\sqrt{5}}{2}\right)^2\overset{?}{\geq}\frac{1+\sqrt{5}}{2}+1$

$1+2\sqrt{5}+5\overset{?}{\geq}\underline{1+\sqrt{5}}+2$

2) $\{n(n+1)\}_{n=1}^{\infty}$ $a_1 = 2$ $a_n = a_{n-1} + 2n$ Compare this to n^2

3) $\{10^n\}_{n=1}^{\infty}$ $a_1 = 10$ $a_n = 10 \cdot a_{n-1}$

Recursively define the set of...

1) even integers $2 \in S$ $x, y \in S \Rightarrow x \pm y \in S$

2) odd integers $1 \in S$ $x, y \in S \Rightarrow x \pm 2y \in S$

3) pos. multiples of 6 $6 \in S$ $x, y \in S \Rightarrow x + y \in S$

Can you describe S without recursion if $1 \in S$ and if $x, y \in S$ then $x+y \in S$ <u>and</u> $x-y \in S$?

Need $\alpha^{k-1} \geq \alpha^{k-2} + \alpha^{k-3} \Rightarrow \alpha^{k-1} \geq (\alpha+1)\alpha^{k-3}$

$\Rightarrow \alpha^2 \overset{?}{\geq} (\alpha+1) \Rightarrow \left(\frac{1+\sqrt{5}}{2}\right)^2 \overset{?}{\geq} \frac{1+\sqrt{5}}{2} + 1$

$\frac{1+2\sqrt{5}+5}{4} \overset{?}{\geq} \frac{1+\sqrt{5}+2}{2}$

$\Rightarrow \frac{6+2\sqrt{5}}{4} \overset{?}{\geq} \frac{3+\sqrt{5}}{2} \Rightarrow \frac{3+\sqrt{5}}{2} \overset{?}{\geq} \frac{3+\sqrt{5}}{2}$ <u>YES</u>.

$\alpha^2 = \alpha + 1$

Notice further that $\alpha^2 = \left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{1+2\sqrt{5}+5}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$ and $\alpha + 1 = \frac{1+\sqrt{5}}{2} + 1 = \frac{3+\sqrt{5}}{2}$ so $\alpha^2 = \alpha + 1$. Then $\alpha^{k-2} + \alpha^{k-3} = (\alpha+1)\alpha^{k-3} = \alpha^2\alpha^{k-3} = \alpha^{k-1}$. So $f_{k+1} < \alpha^{k-1}$ and $P(k+1)$ is true. This completes the inductive step. Therefore by mathematical induction, $f_n < \alpha^{n-2}$ for all integers $n \geq 3$ where $\alpha = \frac{1+\sqrt{5}}{2}$ and f_n is the n^{th} term of the Fibonacci sequence ▨